Riley James Link

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RESEARCH INTERESTS

PhD student in Applied Mathematics at Cornell University researching geometry, optimization, control theory, machine learning, and robotics, with a focus on relaxed symmetry and trajectory tracking.

EDUCATION

Cornell University - Ithaca, New York August 2025 – Expected May 2030

Doctor of Philosophy (PhD) in Applied Mathematics

- *Relevant Coursework:* Lie Groups (in progress), Bayesian Estimation and Stochastic Optimization, Probability I and II, Mathematical Programming I, Continuous Optimization

Creighton University - Omaha, Nebraska August 2021 – May 2025

Bachelor of Science in Mathematics GPA 4.0

- *Relevant Coursework:* Abstract Algebra, Real Analysis, Measure Theory, Data Structures, Computational Physics, Classical Mechanics, Statistical Modeling, Electricity and Magnetism

RESEARCH EXPERIENCE

GeoDesiC Laboratory – Cornell University **January 2026 – Present**

Graduate Researcher – Geometry, Design, and Control Laboratory; working with Jake Welde

- Investigate relaxed Lie symmetries in nonlinear control systems using differential geometry and methods on manifolds. Develop broader theory and explore applications to reducing the dimensionality of trajectory-tracking problems and data-driven dynamics modeling.
- Investigate geometric methods for quadrotor state estimation and control, using the ROS 2 framework.

MITRE – Position, Navigation and Timing: Algorithms and Analysis **May 2024 – August 2024**

Signal Processing Engineering Intern

- Analyzed satellite-based geolocation performance using time- and frequency-difference-of-arrival methods and Cramér–Rao lower bounds.
- Developed a Python codebase for integration into existing repositories and updated a technical report with new performance bounds.

North Carolina State University – DRUMS REU **May 2023 – May 2024**

Undergraduate Researcher

- Developed a framework for approximating perturbed optimal control problems through an NSF- and NSA-funded summer REU; continued the collaboration during the academic year.
- Implemented trajectory optimization methods in MATLAB and coauthored a paper published in SIAM Undergraduate Research Online.

PUBLICATIONS AND PREPRINTS

- Jake Welde, **Riley Link**, and Pieter van Goor. "A Weak Notion of Symmetry for Control Systems." *Preprint*, 2026. arXiv:2609.34173
- **Riley Link**, Pieter van Goor, and Jake Welde. "Reduction of Tracking Control Problems under Weak Symmetry." *IROS Workshop on Geometric Representations in Robotics*, 2026. Non-archival workshop paper. PDF
- **Riley Link** and Ethan Ebbighausen. "A Framework for Approximating Perturbed Optimal Control Problems." *SIAM Undergraduate Research Online*, 18, 2025. doi:10.1137/24S1686000

PRESENTATIONS

- **IROS Workshop on Geometric Representations in Robotics.** *"Reduction of Tracking Control Problems under Weak Symmetry."* Pittsburgh, Pennsylvania, October 1, 2026.
    - Selected for a spotlight talk and received the Best Presentation Award.
- **Joint Mathematics Meetings.** *"A Framework for Approximating Perturbed Optimal Control Problems."* Contributed talk, Session on Research by Undergraduates. Seattle, Washington, January 9, 2025.
- **Joint Mathematics Meetings.** *"A Framework for Approximating Perturbed Optimal Control Problems."* Co-presented at the Undergraduate Research Poster Session. San Francisco, California, January 2024.
- **Iowa MAA Fall Section Meeting.** *"On Approximate Solutions for Perturbed Optimal Control Problems."* Contributed undergraduate talk. Simpson College, Indianola, Iowa, October 28, 2023.
- **NCSU Summer Undergraduate Research and Creativity Symposium.** *"Sensitivity Analysis of Optimal Control Problems."* Poster co-presentation. North Carolina State University, July 27, 2023.

HONORS AND AWARDS

- **Best Presentation Award, IROS Workshop on Geometric Representations in Robotics, 2026.**
- Dean's Award for Distinguished Excellence, Creighton College of Arts and Sciences, 2025; one of four graduating seniors honored.
- Bernard Portz, S.J. Award for Excellence in Mathematics, Creighton University, 2025.
- **Barry Goldwater Scholarship, 2024.**
- Rookie Tutor of the Year, New Horizon Tutoring, 2022.
- Valedictorian, Weiss High School; ranked first in a graduating class of 480.
- Honor societies and programs: Phi Beta Kappa, Pi Mu Epsilon, Alpha Sigma Nu, and Creighton Honors Program.

TEACHING EXPERIENCE

Department of Mathematics – Cornell University **August 2025 – Present**

Teaching Assistant and Grader – MATH 1920: Multivariable Calculus for Engineers

- Served as a teaching assistant and grader for three semesters, leading weekly discussion sections, holding office hours, and providing feedback on homework.

Department of Mathematics – Creighton University **August 2022 – May 2025**

Teaching Assistant and Grader – MTH 350: Applied Linear Algebra and Differential Equations

- Held two 90-minute office-hour sessions each week, graded assignments, and coordinated with the instructor on student progress.

New Horizon Tutoring – North Sioux City, SD **May 2022 – February 2023**

Online Mathematics Tutor

- Prepared lessons and provided individual instruction in high school and early college mathematics, from algebra through Calculus II.

LEADERSHIP SERVICE AND CREATIVE WORK

- Honors Lead Resident Advisor, Creighton University, August 2023 – May 2025. Supported residents, mediated conflicts, and collaborated on orientation activities.
- DREAM Scholars Volunteer, August 2023 – May 2025.
- Consider Initiative Speaker, Creighton University, 2023–2024. Selected to deliver the public talk *Math: Our Silent Ally (and Why It Isn't Terrifying)*.
- Chemistry Department Work Study, Creighton University, August 2021 – May 2025. Assisted with laboratory preparation and departmental operations.
- Wrote and directed a short film titled *Perspective*: https://youtu.be/efpOr8c10rQ